A past–future scalar comparison for a timelike-shell packet

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Abstract

For one registered quadrupole packet in a conditional linear Einstein–scalar shell construction, the complete dynamical boundary problem is even in time, although the initial velocity is nonzero. Comparing the developments with opposite initial velocities removes the initial displacement from a higher graph-energy estimate. This gives the uniform absolute bound $|S(-t) - S(t)| < 264000000001/13738809600 < 20$ for $0 \le t \le 3\ell$, $\ell = 10^{-6}$, where S is the first physical scalar amplitude on the shell in fixed model units. After an explicit state, clock and normalization join to retained future waveform enclosures, two lobe signs first give a coarse past zero. Two selected whole-cell enclosures then localize at least one amplitude zero to $(-6145\ell/4096, -6143\ell/4096)$, a gap of width $\ell/2048$. The result retains the earlier branch, reconstruction, regularity, uniqueness and waveform-certificate premises. It supplies neither a nonlinear amplitude interval nor evidence for a cosmological origin.

1 Registered problem and conditional result

Fix $\ell = 10^{-6}$, $x = t/\ell$, $r = z/\ell$, and the flat unit bump

$$Q(r) = \begin{cases} \exp\left(4 - \dfrac{1}{(r-1)(2-r)}\right), & 1 < r < 2, \\ 0, & \text{otherwise.} \end{cases} \qquad g(r) = \ell e^{-\ell r} Q(r).$$

The registered master data are $U(0, r) = Q(r)$, $U_x(0, r) = g(r)$. The physical first scalar coefficient, including the normal embedding in the retained straight-shell reconstruction, is

$$\Phi_1(\tau, \Omega) = S(\tau/\rho_b) Y(\Omega), \qquad Y = n_1^2 - n_2^2, \quad \tau = \rho_b t. \tag{1}$$

Here $\sup |Y| = 1$ and its angular mean is zero. The background shell scalar is constant, so first clock/label pullbacks add no amplitude term.

We retain the analytic branch, admitted shell enclosures, smooth linear development, full dynamical junction law and physical reconstruction of the past-contact construction [1]. We also identify explicitly when the older actual future waveform enclosure [2] is used. Selected-source hashing and finite arithmetic do not reprove these premises.

Theorem 1. *Under those retained premises, the actual registered first scalar amplitude satisfies*

$$\sup_{0 \le t \le 3\ell} |S(-t) - S(t)| < B, \qquad B = \frac{264000000001}{13738809600} < 20. \tag{2}$$

The compared proper times are $-\rho_b t$ and $+\rho_b t$ at the same anchor. The bound concerns a first linear coefficient in model units.

This is an absolute difference estimate: it does not bound either individual amplitude by 20, prescribe a relative error at a zero, or assert equality of past and future profiles. The construction uses established time-reflection, positive-form and graph-energy ideas [3, 4]; its contribution is the explicit estimate and normalized certificate transfer for this packet.

2 A common operator and a zero-displacement difference

Use the same auxiliary static half-line realization for both developments:

$$\mathcal{H} = L^2(0,\infty) \oplus \mathbb{R}, \qquad \|(u,q)\|_{\mathcal{H}}^2 = \int_0^\infty u^2\,dr + \Lambda q^2, \tag{3}$$

$$A_0(u,q) = \left(-u_{rr} + V_0 u, \frac{-u_r(0) + b_0 q}{\Lambda}\right), \qquad \Lambda = \frac{\mu_b}{\ell} > 1000, \tag{4}$$

$$V_0 = \ell^2 \rho^2 Q_{\mathrm{bg}}, \qquad b_0 = \ell\left(\gamma + \frac{5}{2}\mu_b\right), \quad \gamma = \frac{C_b}{2}.$$

The form domain links $q = u(0)$; the operator domain has no extra static Robin condition. Both endpoint kinetic and spring terms are retained, as in the standard coupled energy bookkeeping [5]. The prior positive-form construction gives $A_0 \geq 5500\ell^2 I$. The complete equation and dynamic boundary law take the common form

$$Z_{xx} + (A_0 + \omega(x)I)Z = 0, \qquad \omega(x) = -\frac{35}{4}\ell^2 \tanh^2(\ell x), \qquad b(x) = b_0 + \Lambda\omega(x). \tag{5}$$

In particular, the boundary spring has the same time dependence as the bulk operator; omitting that endpoint term would invalidate the argument.

The static extension agrees with the actual coefficients through $r = 6$, within the validated background collar $r \leq 8$. The original fields have support within $[0, 2+s]$, and before $s = 1$ within $[1-s, 2+s]$, for $s = |x| \leq 3$. They therefore stay in $r \leq 5$, where the extension has no effect. Their flat initial data satisfy every power-domain compatibility requirement in the retained smooth construction.

Let Z_+ have data $(Q, +g)$ and Z_- data $(Q, -g)$, including the linked zero initial endpoint components. Evenness of the *complete* equation and boundary law, followed by uniqueness, gives

$$Z_{\mathrm{current}}(s) = Z_+(s), \qquad Z_{\mathrm{current}}(-s) = Z_-(s).$$

Thus $\Delta Z = Z_- - Z_+$ has $\Delta Z(0) = 0$, $\Delta Z_s(0) = (-2g, 0)$. The factor two in the velocity is essential. For positive s, $\omega_s \leq 0$, so the complete estimating energy

$$E[X] = \frac{1}{2}\left(\|X_s\|_{\mathcal{H}}^2 + \|A_0^{1/2}X\|_{\mathcal{H}}^2 + \omega\|X\|_{\mathcal{H}}^2\right), \qquad E'[X] = \frac{\omega_s}{2}\|X\|_{\mathcal{H}}^2 \leq 0 \tag{6}$$

is nonincreasing. It is a mathematical norm estimate, with no physical injected-energy or Noether-energy interpretation supplied here. Since $\|Q\|_2 \leq 1$, $E[\Delta Z](s) \leq E[\Delta Z](0) \leq 2\ell^2$. Its local endpoint trace Δq vanishes before contact. From the endpoint mass in (6), $500|\Delta q_s|^2 \leq E[\Delta Z](0)$, whence

$$|\Delta q(s)|^2 \leq \frac{(s-1)_+^2}{500} E[\Delta Z](0) \leq \frac{2\ell^2}{125} < \ell^2, \qquad 0 \leq s \leq 3. \tag{7}$$

3 Graph energy and the physical scalar bound

Apply the abstract form identity to $A_0^{1/2}\Delta Z$. This vector is generally nonlocal; no compact-support assertion is made for it. Scalar ωI commutes with A_0, and coercivity gives

$$2E[A_0^{1/2}\Delta Z](s) \geq (1-\epsilon)\|A_0 \Delta Z(s)\|_{\mathcal{H}}^2, \qquad \epsilon = \frac{63\ell^2}{4400} < 1, \tag{8}$$

$$2E[A_0^{1/2}\Delta Z](0) = 4a_0(g,g) \leq 4\ell^2\left[(16+\ell)^2 + 103000\ell^2\right]. \tag{9}$$

To see the constants, $|\omega| \leq (315/4)\ell^4$ for $s \leq 3$, and $V_0 < 103000\ell^2$ on the initial support. Writing $y = [(r-1)(2-r)]^{-1} \geq 4$ gives $Q_r = (3-2r)y^2 e^{4-y}$, so $|Q_r| \leq 16$ analytically and $\|g_r\|_2 \leq \ell(16+\ell)$. The zero displacement removes the initial Q_{rr} contribution. The spectral identity in (9) includes every endpoint component of the half-power vector. Exact arithmetic yields

$$\|A_0 \Delta Z(s)\|_{\mathcal{H}}^2 \leq \frac{4\ell^2[(16+\ell)^2 + 103000\ell^2]}{1-\epsilon} < (33\ell)^2. \tag{10}$$

The retained physical reconstruction and boundary cancellation give

$$S_\Delta(\ell s) = \frac{3\mu_b \cosh(\ell s)^{-3/2}}{\rho_b^3 p_b}\left[-\frac{(A_0\Delta Z)_b}{\ell^2} + \frac{5}{2}\Delta q\right]. \tag{11}$$

Indeed $U_r = b_0 q - \Lambda(A_0 Z)_b$ implies $U_r/\ell - \gamma q = \mu_b[-(A_0 Z)_b/\ell^2 + (5/2)q]$. The temporal term cancels. The normal derivative is spatial and the conjugating cosh factor is even, so there is no additional reflection minus sign: $S_\Delta(t) = S(-t) - S(t)$.

The endpoint norm implies $\mu_b|(A_0\Delta Z)_b| \le \sqrt{\mu_b\ell}\|A_0\Delta Z\|_{\mathcal{H}}$. Use the admitted bounds $\mu_b < 1/400$, $\sqrt{\mu_b\ell} < 1/20000$, $\rho_b \ge 129$, $|p_b| \ge 3/25000$, together with (7) and (10). Then

$$|S_\Delta| < \frac{25000}{129^3}\left(\frac{33}{20000\ell} + \frac{\ell}{160}\right) = \frac{264000000001}{13738809600} = B < 20. \tag{12}$$

This proves Theorem 1. It also bounds the angular first-field difference because $|Y| \le 1$, without supplying an SI calibration.

4 A retained future certificate and a coarse past zero

The selected wider inertial waveform proof [2] uses this same $\ell = 10^{-6}$, branch, flat bump, nonzero velocity $e^{-z}Q(z/\ell)$, quadrupole, conjugation and full dynamic shell law. Its actual enclosure includes the proved error from the free comparator. The report exports

$$H(x) = \ell\,[w_z(\ell x, 0) - \gamma w(\ell x, 0)], \qquad \ell S(\ell x) = \frac{3\cosh(\ell x)^{-3/2}}{\rho_b^3 p_b} H(x). \tag{13}$$

Thus the stored `ell_S` interval must be divided by ℓ *before* adding the model-unit difference ceiling B. The same reconstructed state and clock are identified by retained linear uniqueness; neither a comparator alone nor the separate $\ell = 10^{-10}$ packet supplies this join.

If $[L, U]$ encloses the actual future value $S(t)$, then $[L - B, U + B]$ encloses the matched past value $S(-t)$. Table 1 shows outward decimal roundings of the exact rational endpoints. The proof uses the rational intervals, not those readable decimal summaries.

Matched $t > 0$	Retained future $S(t)$	Transported past $S(-t)$
$5\ell/4$	$[-41241.437, -37396.209]$	$[-41260.653, -37376.993]$
$7\ell/4$	$[37398.637, 41244.115]$	$[37379.422, 41263.331]$

Table 1: Two actual coefficient enclosures in model units, conditional on the retained future enclosure theorem and Theorem 1.

Corollary 2. *With the retained future certificate, $S(-5\ell/4) < -37376$ and $S(-7\ell/4) > 37379$. The continuous amplitude therefore has at least one zero in $(-7\ell/4, -5\ell/4)$.*

A selected-cell refinement. The older certificate also supplies continuous enclosures on whole closed grid cells, rather than only sampled values. Authenticating its cell CSV and retaining the earlier full-cell enclosure theorem gives the two uniform transports

$$\begin{array}{l|l} s \in [3071/2048,\ 6143/4096] & S(-\ell s) < -53 \\ s \in [6145/4096,\ 3073/2048] & S(-\ell s) > 63. \end{array} \tag{14}$$

The CSV stores integer `ell_S` endpoints with value denominator $D_v = 2^{112}$. Decode them by D_v, then divide by ℓ to obtain physical-S endpoints and widen by B. No old waveform campaign or numerical PDE is rerun. Applying continuity to the two inner cell endpoints proves the sharper conditional statement

$$\exists\, t_0 \in \left(-\frac{6145\ell}{4096}, -\frac{6143\ell}{4096}\right) :\ S(t_0) = 0, \qquad \text{gap width} = \frac{\ell}{2048}. \tag{15}$$

This gap is exactly 1024 times narrower than the preceding coarse past interval of width $\ell/2$. This comparison concerns those two past intervals only; it does not compare precision with older fine future brackets or assert optimal localization.

Neither zero conclusion gives uniqueness or a total zero count. These are signs of the coefficient S, rather than a globally signed angular field. The four older narrow central/late point witnesses and negative future tail have margins smaller than B; their widened intervals straddle zero. Those particular older point-witness brackets and the tail sign therefore do not transfer by this ceiling. The selected-cell argument establishes a different past interval. A quadratic response integral needs a separate norm and cross-term argument, rather than pointwise interval widening.

Scope and reproducibility. Selected local input copies, a continuous primary proof, an independent odd-part derivation, a separately checked whole-cell certificate join, exact normal/optimized arithmetic reports and the manuscript display audit accompany this note in the research capsule. The retained older waveform theorem is accepted as a conditional premise; its full campaign is not rerun. The new calculations use rational arithmetic and source checks, with no numerical PDE or field/source/bump oracle. Internal review is distinct from external peer review and formal proof-assistant certification.

Earlier future-contact work [6] already constructs a conditional complete finite-angular second Taylor coefficient under its correction, gauge, corner and support hypotheses. This note concerns the first coefficient. Quantitative control of the chosen past second-order observable and particular baseline, a nonlinear defect-to-error map and effective parameter interval, physical incoming preparation, radiation and cosmological-origin evidence remain separate tasks.

References

[1] R. Maldonado, *A pointwise scalar-trace bound through past shell contact*, working paper and source capsule (4 November 2026), HDBLAST past-shell-contact checkpoint.

[2] R. Maldonado, *Inertial shell waveform*, primary proof, actual enclosure report and selected source inputs (15 October 2026), HDBLAST inertial-waveform checkpoint.

[3] J. Zahn, *Generalized Wentzell boundary conditions and quantum field theory*, Annales Henri Poincaré **19**, 163 (2018), doi:10.1007/s00023-017-0629-3; arXiv:1512.05512.

[4] A. Marta, *The nonlinear wave equation with nonlinear Wentzell boundary conditions on time-dependent compact Riemannian manifolds*, arXiv:2404.05855 (2024), especially Remark 4.3 on reversal of time and Remarks 4.4–4.5 on the coupled energy and finite speed.

[5] A. Dahmani, Y. Chitour, H.-M. Nguyen and C. Roman, *One dimensional wave equation with in-domain localized damping and Wentzell boundary conditions*, arXiv:2503.22439 (2025), Section 4, Eqs. (4.2)–(4.3). Its damping-dependent decay estimate is not imported here.

[6] R. Maldonado, *A complete finite-angular second-order coefficient through shell contact* (14 October 2026), HDBLAST full-second-order-contact primary proof, Sections 1 and 8.